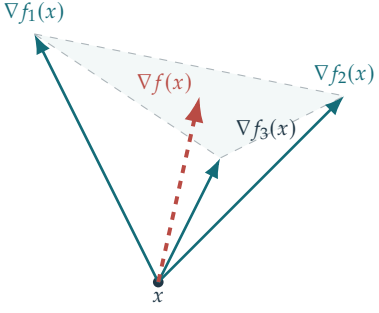
Panel 1

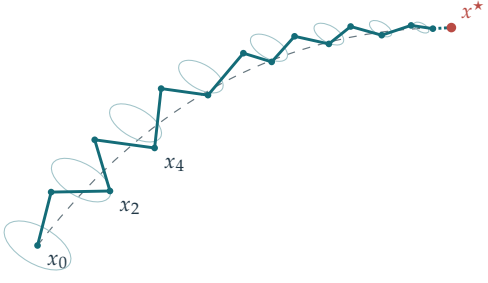
$$\mathbb{E}_i[\nabla f_i(x)] = \nabla f(x)$$



$f = \frac{1}{3}(f_1 + f_2 + f_3)$

$i \sim \text{Unif}\{1, 2, 3\}$

Panel 2



$$x_{k+1} = x_k - \tau_k \nabla f_{i(k)}(x_k)$$

Illustrative trajectory
and shrinking neighborhoods